MATH-101 Calculus and Analytical Geometry – 3-0 CHs

By

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INDETERMINATE FORMS

Lec # 7

OBJECTIVE

- In this lecture we will discuss a general method for using derivatives to find limits.

- The method that we will discuss in this lecture is an extremely powerful tool that is used internally by many computer programs to calculate limits of various types.

INDETERMINATE FORMS
&
L' HOSPITAL'S RULE

The Form $\left(\frac{0}{0}\right)$ or $\left(\frac{\infty}{\infty}\right)$

Definition:

1) If $lim_{x\to a}\, f(x) = 0$ and $lim_{x\to a}\, g(x) = 0$ then $\frac{f(x)}{g(x)}$ is called an indeterminate form at $x = a$, of the type $\left(\frac{0}{0}\right)$.

2) If $lim_{x\to a}\, f(x) = \pm\infty$ and $lim_{x\to a}\, g(x) = \pm\infty$ then $\frac{f(x)}{g(x)}$ is called an indeterminate form at $x = a$, of the type $\left(\frac{\infty}{\infty}\right)$.

THEOREM L' HOPITAL'S RULE

Suppose $'f(x)'$ and $'g(x)'$ are differentiable and $g'(x) \neq 0$, near 'a' (except possibly at a).

Suppose that

$$lim_{x \to a}\, f(x) = 0 \text{ and } lim_{x \to a}\, g(x) = 0$$

or

$$lim_{x \to a}\, f(x) = \pm\infty \text{ and } lim_{x \to a}\, g(x) = \pm\infty$$

[In other words, we have an indeterminate form of type $\left(\frac{0}{0}\right)$ or $\left(\frac{\infty}{\infty}\right)$] Then

$$lim_{x \to a}\, \frac{f(x)}{g(x)} = lim_{x \to a}\, \frac{f'(x)}{g'(x)}$$

Example Find $lim_{x\to 1}\dfrac{\ln(x)}{x-1}$

Solution:

Since $lim_{x\to 1}\ln(x) = \ln(1) = 0$ and

$$lim_{x\to 1}(x-1) = (1-1) = 0 \quad \text{so} \quad \left(\frac{0}{0}\right)$$

We can apply L' Hospital's Rule

$$lim_{x\to 1}\frac{\ln(x)}{x-1} = lim_{x\to 1}\left(\frac{\frac{d}{dx}(lnx)}{\frac{d}{dx}(x-1)}\right)$$

$$= lim_{x\to 1}\left(\frac{\frac{1}{x}}{1}\right) = lim_{x\to 1}\left(\frac{1}{x}\right) = 1$$

Example Find $lim_{x \to \infty} \left(\frac{e^x}{x^2}\right)$

Solution:

$$lim_{x \to \infty} \left(\frac{e^x}{x^2}\right) = lim_{x \to \infty} \left(\frac{e^x}{x^2}\right) = \left(\frac{\infty}{\infty}\right)$$

$$= lim_{x \to \infty} \left(\frac{e^x}{2\,x}\right) \quad \left(\frac{\infty}{\infty}\right) \quad \text{(By L' Hospital's Rule)}$$

$$= lim_{x \to \infty} \left(\frac{e^x}{2}\right) \quad \text{(By L' Hospital's Rule)}$$

$$= \infty$$

Example: Calculate $lim_{x \to \infty} \left(\frac{ln\, x}{\sqrt[3]{x}} \right)$

Solution:

Since $ln\, x \to \infty$ and $\sqrt[3]{x} \to \infty$ as $x \to \infty$.

So,

$$= lim_{x \to \infty} \left(\frac{ln\, x}{\sqrt[3]{x}} \right)$$

$$= lim_{x \to \infty} \left(\frac{1/x}{\frac{1}{3} x^{-2/3}} \right)$$ (By L' Hospital's Rule)

$$= lim_{x \to \infty} \left(\frac{3}{\sqrt[3]{x}} \right) = 0$$

THE FORM

$(0 \cdot \infty)$ or $(\infty \cdot 0)$

THEOREM: Evaluate

$lim_{x \to a} [f(x) \cdot g(x)]$ when $lim_{x \to a} f(x) = 0$ and

$$lim_{x \to a} g(x) = \infty$$

We write

$$f(x) \cdot g(x) = \frac{f(x)}{1/g(x)} \qquad \left(\frac{0}{0}\right)$$

or

$$= \frac{g(x)}{1/f(x)} \qquad \left(\frac{\infty}{\infty}\right)$$

So that limits of these forms can be calculated by the previous method.

Example

Find $\lim_{x \to \infty} x\left(a^{1/x} - 1\right), \quad a > 0$

Solution:

$$\lim_{x \to \infty} x\left(a^{1/x} - 1\right) \qquad (\infty \cdot 0)$$

$$= \lim_{x \to \infty} \frac{\left(a^{1/x} - 1\right)}{1/x} \qquad \left(\frac{0}{0}\right)$$

$$= \lim_{x \to \infty} \frac{a^{1/x} \ln(a) \left(-1/x^2\right)}{\left(-1/x^2\right)}$$ (By L' Hospital's Rule)

$$= \lim_{x \to \infty} a^{1/x} \ln(a) = \ln(a)$$

THE FORM $(\infty - \infty)$

THEOREM:

$$\lim_{x \to a} [f(x) - g(x)]$$

when

$$\lim_{x \to a} f(x) = \infty = \lim_{x \to a} g(x).$$

We can write

$$f(x) - g(x) = \frac{1/g(x) - 1/f(x)}{1/f(x)g(x)}$$

The R.H.S. is of the form $\left(\frac{0}{0}\right)$ and can be evaluated by L'Hospital Rule.

EXAMPLE Evaluate $lim_{x \to 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$

Solution:

$$= lim_{x \to 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) \qquad (\infty - \infty)$$

$$= lim_{x \to 0} \left(\frac{x - \sin x}{x \sin x} \right) \qquad \left(\frac{0}{0} \right)$$

$$= lim_{x \to 0} \left(\frac{1 - \cos x}{\sin x + x \cos x} \right) \qquad \text{(By L' Hospital's Rule)}$$

$$= lim_{x \to 0} \left(\frac{\sin x}{2 \cos x - x \sin x} \right) = \frac{0}{2} = 0$$

THE FORMS $(0^0,\ 1^\infty,\ \infty^0)$

THEOREM:

$$\lim_{x \to a} [f(x)]^{g(x)} \quad \text{when}$$

1. $\lim_{x \to a} f(x) = 0 \qquad \lim_{x \to a} g(x) = 0.$
2. $\lim_{x \to a} f(x) = 1 \qquad \lim_{x \to a} g(x) = \infty.$
3. $\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a} g(x) = 0.$

Here we let

$$y = [f(x)]^{g(x)}, \text{ then}$$

$$lny = g(x)ln\big(f(x)\big) \longrightarrow \text{Eq. (1)}$$

(using log properties)

THE FORMS $(0^0, 1^\infty, \infty^0)$ (*cont.*)

1) If $\lim_{x \to a} f(x) = 0 = \lim_{x \to a} g(x)$, then (1) is of the form $(0 \cdot \infty)$.
2) If $\lim_{x \to a} f(x) = 1$, $\lim_{x \to a} g(x) = \infty$, then (1) is again of the form $(\infty \cdot 0)$.
3) If $\lim_{x \to a} f(x) = \infty$, $\lim_{x \to a} g(x) = 0$, then (1) is of the form $(0 \cdot \infty)$.

Thus all the limits can be evaluated from the form $0 \cdot \infty$.

Suppose

$$\lim_{x \to a}[g(x)\, ln\ (f(x))] = L$$

Then,

$$\lim_{x \to a} lny = L \qquad \text{from (1)}$$

or

$$ln\,(\lim_{x \to a} y) = L \quad \text{i.e.} \quad \lim_{x \to a} y = e^{L}$$

Example Evaluate $\lim_{x \to 0} x^{x}$

Solution: Let $y = x^{x}$

$$\Rightarrow \ln y = \ln(x^{x}) = x \ln x$$

Now $\lim_{x \to 0} \ln y = \lim_{x \to 0} x \ln x$ ⟶ Eq. (1)

Now

$$\lim_{x \to 0} x \ln x = \lim_{x \to 0} x \ln x \qquad (0 . \infty)$$

$$= \lim_{x \to 0} \left(\frac{\ln x}{1/x} \right) \qquad \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \to 0} \left(\frac{1/x}{-1/x^{2}} \right) = \lim_{x \to 0} (-x) = 0$$

So, eq(1) becomes

$$\lim_{x \to 0} \ln y = 0 \implies \lim_{x \to 0} y = e^{0} = 1$$

$$\implies \lim_{x \to 0} x^{x} = 1$$

Exercise

Evalute the given limits.

1) $\lim_{x \to 0} \dfrac{\tan x - x}{x^3}$ Ans $\longrightarrow$ $1/3$

2) $\lim_{x \to 0} \left(\dfrac{1}{x} - \cot x \right)$ Ans $\longrightarrow$ 0.

3) $\lim_{x \to \infty} \dfrac{x - 2x^2}{3x^2 + 5x}$ Ans $\longrightarrow$ $-2/3$

4) $\lim_{x \to 0} \dfrac{1 - \cos x}{x + x^2}$ Ans $\longrightarrow$ $1/2$.

5) $\lim_{x \to 0} \dfrac{x \cos x - \ln(1+x)}{x^2}$ Ans $\longrightarrow$ $1/2$

6) $\lim_{x \to \pi/2} \dfrac{\tan x}{\tan 3x}$ Ans $\longrightarrow$ 3

7) $\lim_{x \to 0} \dfrac{\sqrt{9+5x+4x^2} - 3}{x}$ Ans $\longrightarrow$ $5/6$.

8) $\lim_{x \to 0} \dfrac{\ln(1 + a \sin x)}{\sin x}$ $\xrightarrow{\text{Ans}}$ a

9) $\lim_{x \to a} \left[\sin(x-a) \right]^{(x-a)}$ $\xrightarrow{\text{Ans}}$ 1

10) $\lim_{x \to 0} \left(\frac{1+\tan x}{1+\sin x} \right)^{1/\sin x}$ $\xrightarrow{\text{Ans}}$ 1

11) $\lim_{x \to 1} \frac{1-x+\ln x}{1-\sqrt{2x-x^2}}$ $\xrightarrow{\text{Ans}}$ -1

12) $\lim_{x \to a} (x-a) \operatorname{Cosec}\left(\frac{\pi x}{a}\right)$ $\xrightarrow{\text{Ans}}$ $-a/\pi$

13) $\lim_{x \to \pi/2} (\tan x)(\ln \sin x)$ $\xrightarrow{\text{Ans}}$ 0

14) $\lim_{x \to 1} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$ $\xrightarrow{\text{Ans}}$ $1/2$

15) $\lim_{x \to \infty} \left(\sqrt{x^2 + 5x} - x \right)$ $\xrightarrow{\text{Ans}}$ $5/2$

16) $\lim_{x \to 1} x^{\frac{1}{1-x}}$ $\xrightarrow{\text{Ans}}$ $1/e$.